

Ly Panharith

Sangkat Teuk Thla, Khan Sen Sok, Phnom Penh, Cambodia

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GitHub | LinkedIn

Education

Royal University of Phnom Penh <i>Bachelor of Science, Data Science and Engineering — 3rd Year</i>	2023 – Present
Institute of Foreign Languages, Royal University of Phnom Penh <i>Bachelor of Arts, English — 3rd Year</i>	2023 – Present
Hun Sen Mittapheap High School <i>High School Diploma — Baccalaureate II, Grade A</i>	2023

Experience

Artificial Intelligence Engineer Intern <i>Ministry of Industry, Science, Technology & Innovation (MISTI)</i>	Oct 2025 – Present
- Develop computer vision models for CCTV anomaly detection using real-world video footage. - Apply deep learning and object detection techniques for real-time video analysis, model evaluation, and iterative performance improvement. - Contribute to an AI-powered security monitoring system integrating computer vision with a web-based monitoring platform and automated alerts.	

Volunteer Experience

Translator <i>Singapore Polytechnic — OSIP 2026</i>	Mar – Apr 2026
Facilitated communication between Singaporean students and pepper farmers in Kampot during field research, interviews, and prototype development.	
Project Member <i>KimiTABI</i>	2026 - Present
Collaborated with Cambodian and Japanese students on community-based projects, contributing to cross-cultural communication, teamwork, and social impact initiatives.	
Logistics Coordinator <i>VMC: Volunteer for My Community</i>	Dec 2025
Coordinated logistics, prepared resources, and supported team operations to ensure smooth delivery of community engagement activities.	

Honors and Awards

2024 Techo Digital Talent Scholarship <i>Ministry of Post and Telecommunications</i>	2024
Awarded a scholarship to pursue a Bachelor's degree in Data Science at the Royal University of Phnom Penh in recognition of academic merit and potential in digital technology.	
2nd Award, Top Project <i>Water Misting System — Faculty of Engineering, RUPP</i>	2024
Awarded Second Place at the 7th Engineering Day Awards 2024 for the development of a water misting system.	
Baccalaureate II, Grade A <i>Hun Sen Mittapheap High School</i>	2023
Achieved Grade A in Cambodia's national high school examination.	

Projects

SecureVision AI Surveillance

- Developed an enterprise-style CCTV monitoring platform using YOLO object detection and a Django back-end for real-time threat detection, motion tracking, and automated security alerts.
- *Technologies:* Python, YOLO, Django, OpenCV, MySQL
- GitHub Repository

AI Agent for University Room Reservation

- Developed an intelligent university room-booking system using Agentic RAG and Semantic Kernel orchestration, with integration of face recognition, chatbot interaction, mobile access, calendar synchronization, and containerized deployment.
- *Technologies:* Python, Django, LangChain, Semantic Kernel, ChromaDB, Flutter, Docker, MySQL
- GitHub Repository

Diabetes Prediction System

- Developed a binary classification model for diabetes prediction using the TAIPEI dataset containing approximately 15,000 patient health records.
- Built and deployed an interactive web application for model-based prediction and visualization.
- *Technologies:* Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit
- GitHub Repository | Live Demo

Sleep & Academic Concentration Research

- Conducted a quantitative study examining the relationship between sleep deprivation and academic concentration among undergraduate students.
- Performed data cleaning, reliability analysis, correlation analysis, independent-samples t-tests, regression analysis, and data visualization.
- *Technologies:* SPSS, Statistics, LaTeX, Excel
- Research Project

Certifications

- AWS Academy Machine Learning Foundations
- Essentials of AI Learning Frameworks & Advanced Models
- SAP Analytics Cloud Training
- Dynamic Entrepreneurial Course

Technical Skills

Programming: Python, C++, SQL, JavaScript, HTML, CSS

Machine Learning & Data Science: Supervised Learning, Unsupervised Learning, Deep Learning, Data Analysis, Statistical Analysis, Scikit-learn

AI & LLM: YOLO, OpenCV, RAG, Agentic AI, LangChain, Semantic Kernel, ChromaDB

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, SPSS

Web & Database: Django, Flask, Streamlit, MySQL

Tools: Git, GitHub, Docker, Jupyter Notebook

Languages:

Khmer (Native), English (Upper-Intermediate)

Interpersonal Skills:

Conflict Resolution, Problem Solving, Teamwork, Interpersonal Communication, Time Management, Patience